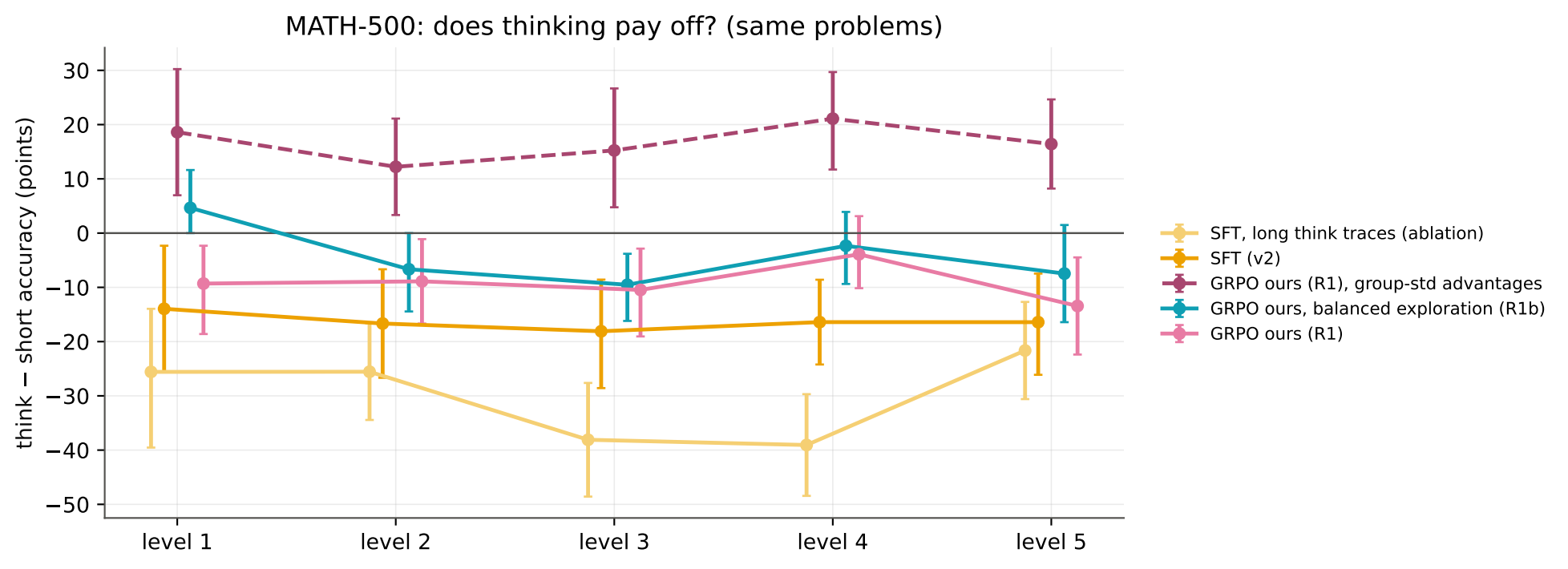
MATH-500: does thinking pay off? (same problems)
think − short accuracy (points)
30
20
10
0
−10
−20
−30
−40
−50
level 1
level 2
level 3
level 4
level 5
SFT, long think traces (ablation)
SFT (v2)
GRPO ours (R1), group-std advantages
GRPO ours, balanced exploration (R1b)
GRPO ours (R1)